

# Ethan Corr

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## Education

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### Marquette University

- PhD in Computational Mathematics & Statistical Sciences Expected May 2029
- Advisor: Dr. Gregory Ongie
  - Dissertation: “An Analytical Approach to Neural Net Behavior”
- MS in Computational Mathematics & Statistical Sciences May 2025
- MS Essay: “Characterization of 1D Interpolating Neural Networks”
- BS in Mechanical Engineering, Mathematics May 2020
- University Honors Program
  - Summa Cum Laude

## Honors & Awards

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- Discovery World Hackathon Winning Team** Feb 2026  
Northwestern Mutual Data Science Institute, Discovery World
- Outstanding Mathematics Senior Award** May 2020  
Marquette University

## Professional Experience

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**Data Scientist** Dec 2025 – Present  
*Clarios* Glendale, WI

- Develop end-to-end supply chain network optimization models for supply planning business users
  - Implement linear programming model to allocate supply with minimum cost, saving +\$2M annually
  - Integrate AI agents to augment human user edits and scenario running capabilities, as well as support explainability of results for business users
- Forecast customer core return rates by region using seasonal weather features and customer POS data

**AI/ML Algorithm Engineer** July 2023 – Dec 2025  
*Clarios* Glendale, WI

- Developed forecasting and prediction models for battery state estimation using ensemble methods.
  - State-of-Charge (SOC) forecast model using random forest classifier and LSTM network
  - Battery cranking health model using vehicle classification and prior crank signal distribution
- Implemented battery cranking health and replacement algorithm in SnowPark data warehouse (Snowflake)
- Led design and validation of “Battery Bot” GenAI personal productivity tool for Clarios engineering
  - Tested and evaluated custom-built solutions using OpenAI models with Python API, local embedding methods using Ollama RAG models, and low-code agentic model framework in Copilot Studio

**Design Engineer** June 2020 – July 2023  
*Clarios* Glendale, WI

- Developed Python tool to analyze tolerance stacks of hole patterns using statistical methods
- Designed battery packaging around Li-ion cells per thermal management and cell protection requirements
- Conducted part-level mechanical simulations and coordinated assembly-level simulations with offshore team

## Teaching Experience

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### K12 STEM Outreach Lead

July 2021 – July 2025

*Clarios*

*Glendale, WI*

- Led “Engineering Challenge Day” four-time annual event at MSOE STEM Center, a half-day event for middle school students to do four different STEM challenges and learn about being a STEM professional
- Developed team competencies in conference presentations, communications, and collecting student feedback

### Teaching Assistant

Aug 2019 – Mar 2020

*Marquette University (Dept. of Mechanical Eng.)*

*Milwaukee, WI*

- Instructed undergraduate engineering students in engineering measurements & controls lab activities

### Mathematics Peer Tutor

Aug 2017 – May 2019

*Marquette University (Office of Student Ed. Services)*

*Milwaukee, WI*

- Demonstrated problem solving methods to groups of 4-6 students in calculus and differential equations

## Grants & Funding Awards

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### NMDSI Student Scholars - Fall 2026

August 2026

Northwestern Mutual Data Science Institute award to study Beurling Lasso solution path algorithm

### NMDSI Student Scholars - Summer 2026

April 2026

Northwestern Mutual Data Science Institute award to study Beurling Lasso solution uniqueness

## Mentorship

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**Benjie Harthun** (Undergraduate Intern, Clarios)

June 2026 – Present

LP for supply chain optimization, data pipelines for ERP systems

**Sri Vishal Payaniappa** (Undergraduate Intern, Clarios)

June 2025 – Jan 2026

RAG-enabled GenAI for engineering requirements analysis, process automation

**Prathmesh Konda Gopinath** (Undergraduate Intern, Clarios)

May 2024 – Jan 2025

Data analysis to characterize battery-vehicle behavior, voltage signal processing

## Conference Presentations

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“Masters of the Mundane: Unveiling STEM Potential through Everyday Expertise”

October 2025

STEM Forward, 2025 SySTEMnow Conference. Milwaukee, WI

## General Talks & Presentations

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“The Battery Replacement Problem”

October 2025

Marquette University, MSSC Department Colloquium

“Characterization of 1D Interpolating Neural Networks”

April 2025

Marquette University, Master’s Essay Presentation

“Battery Bot: Engineering Powered by AI”

January 2025

Clarios, Americas Regional Tech Talk

“Equivalent Circuit Models & Mathematics in Industry”

November 2024

Marquette University, Student Success Seminar

## Conferences Attended

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2026 Data-Driven Wisconsin

August 2026

Milwaukee School of Engineering; Milwaukee, WI

2025 Deep Learning Workshop

March 2025

Marquette University; Milwaukee, WI

## Patents

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### Temperature Sensor for a Battery

*Pending*

Mechanical apparatus to affix temperature sensor in cell stack

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|-----------------------------------------------------------------------------|----------------|
| <b>Customizable Battery</b>                                                 | <i>Pending</i> |
| Modular battery design for multiple cell configurations                     |                |
| <b>Cell swell restrictor and cell stack fixation</b>                        | <i>Pending</i> |
| Cell swell restrictor (CSR) configured to protect a battery                 |                |
| <b>Crush and swelling protection of battery components</b>                  | <i>Pending</i> |
| A battery protection system to protect a battery with a cell housing        |                |
| <b>Battery management system structure and battery comprising same</b>      | <i>Pending</i> |
| Integrated BMS connector for automotive Li-ion battery application          |                |
| <b>Interface for Vehicle Battery</b>                                        | October 2025   |
| USD1096635S1. Ornamental design of vehicle-battery interface                |                |
| <b>Vehicle Battery</b>                                                      | October 2025   |
| USD1096612S1. Ornamental design of Li-ion automotive battery system vehicle |                |